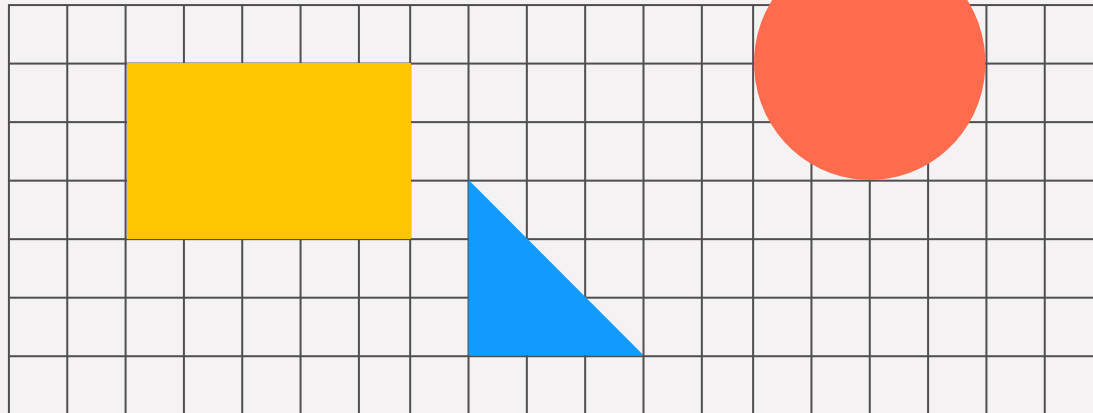


Median Income and Educational Outcomes in Texas



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Guiding Research Questions



- What is the relationship between median household income and educational outcomes in Texas?
- Across all counties, what is the range of median household income?
- What are the trends in educational outcomes in Texas across the counties?
 - **Educational outcomes include: graduation rates, college readiness, and attendance**

Data Wrangling

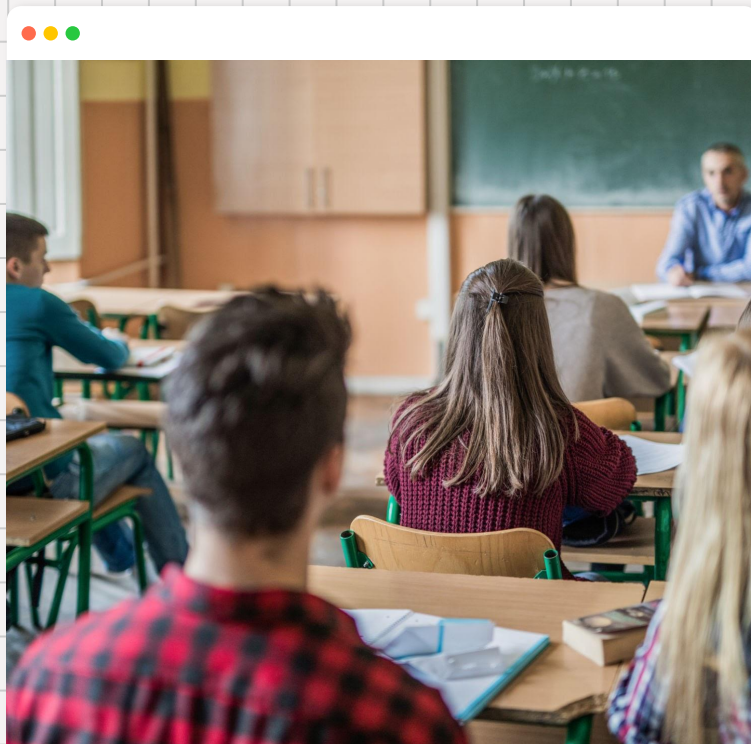


Collect data from Texas education agency

Filter out missing data points from schools/counties

Imported Census data for Texas income

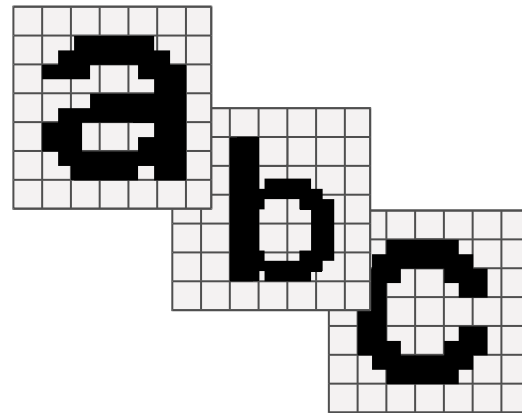
**Collected Texas shapefile for Texas map
Align the names of Texas counties**



Preliminary Observations

Observations:

- Attendance rates do not significantly vary from year to year
- Generally, there is a relationship between greater household income average college readiness



Resources

- Plotly
- Ggplot
- Readr
- Tidyverse
- Gganimate

Texas Median Income by County: {frame@time}

36 ° N

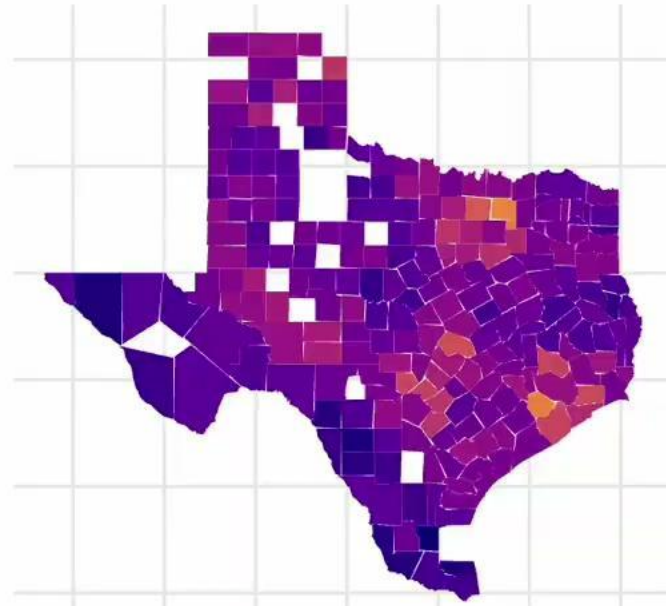
34 ° N

32 ° N

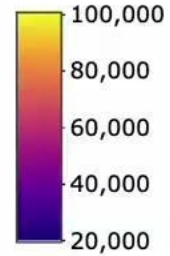
30 ° N

28 ° N

26 ° N



Median Income



106 ° W 104 ° W 102 ° W 100 ° W 98 ° W 96 ° W 94 ° W

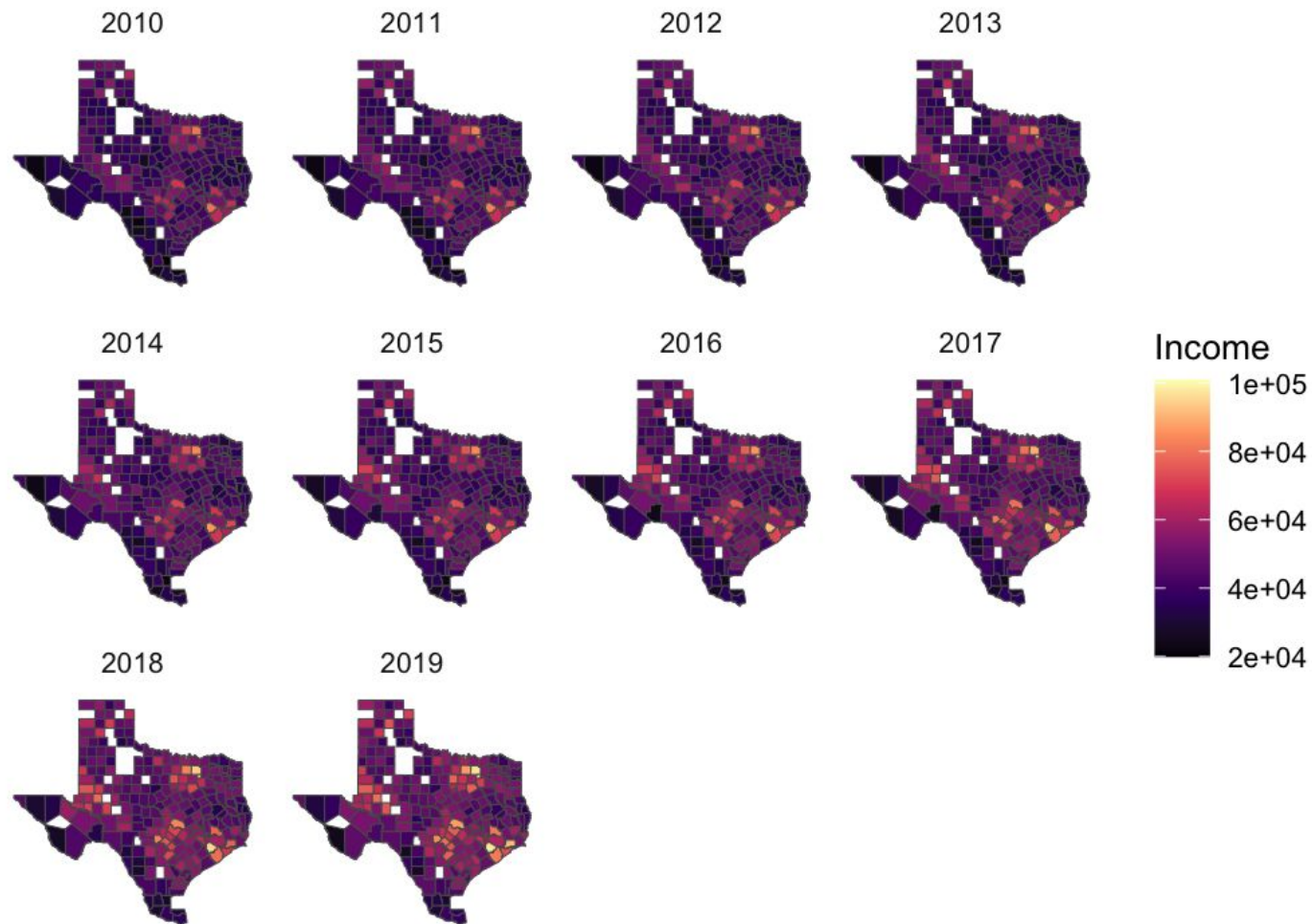
Year: 2010

Play Animation

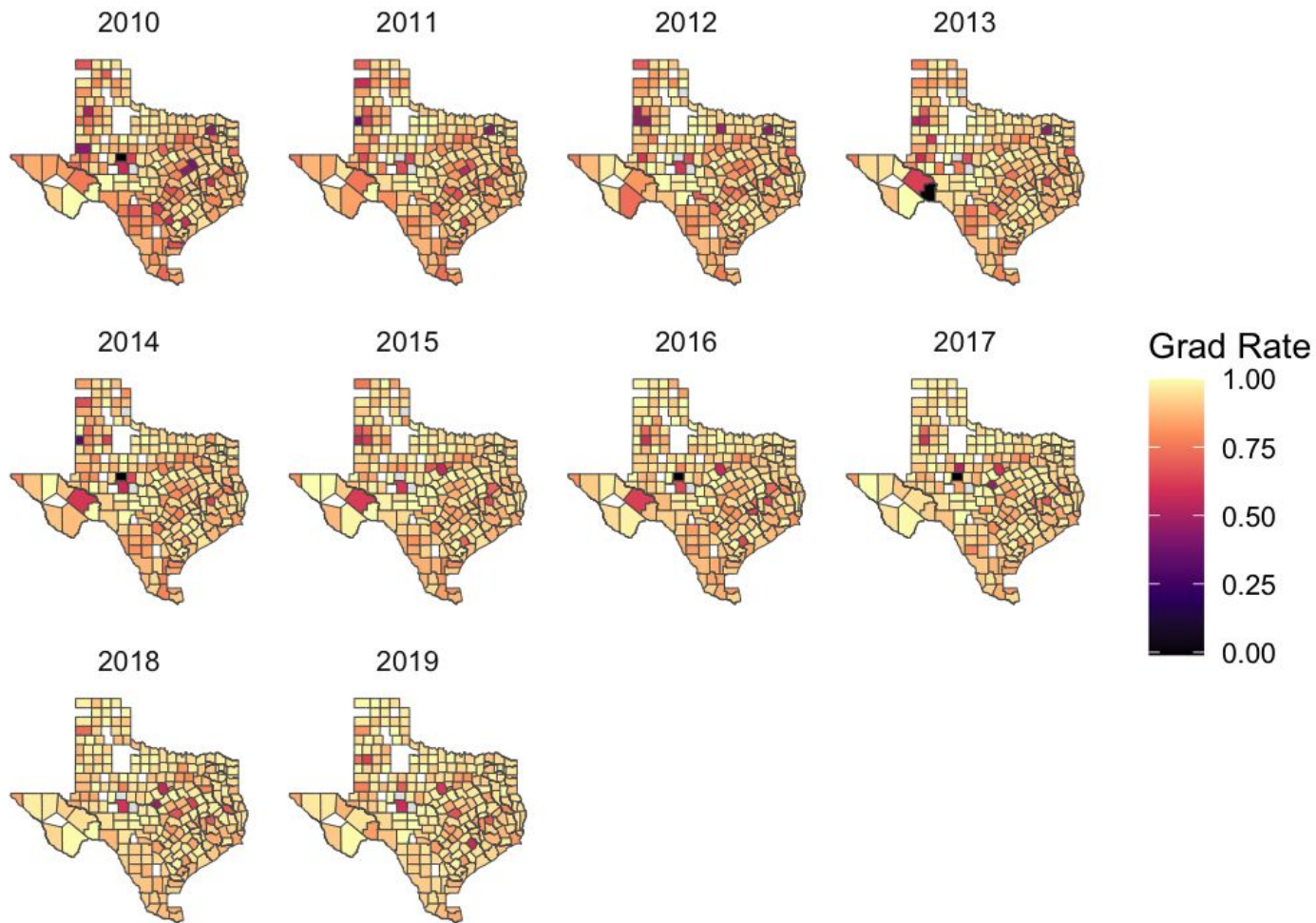


2010 2011 2012 2013 2014 2015 2016 2017 2018 2019

Income by County in Texas (by Year)



Four-Year Graduation Rate by County in Texas (by Year)



Final Visualizations

Goal: We wanted to create a sliding visualization of maps of Texas that show different parts of our data.

- **We see an increase in average household income over the span of the ten years we visualize**
 - This makes sense because of national economic trends
- **We can see the general trend that students from on average higher household incomes are more college ready (based on test scores) than students who come from on average lower income areas.**
 - More families from mid economic range than higher or lower ranges.

These are just trends and not predictive of all college readiness across the state.

Note

Limitations

- There are missing counties in our visualizations
 - Some county info is just missing from the data
 - We removed counties that miss observations from the years we look at
- Educational outcomes have selection bias
- How college readiness is calculated changed, making the data useless

Thank you!

